MGMT590

Big Data and MLOps Final Project Report

Annabelle Harris, Mia Kim, Daniel Kang,
and Shubh Vaishnav

Phase 1: Data Selection & Problem Definition

This project analyzes a structured marketing dataset from three major Indian beauty cosmetics and e-commerce brands, Nykaa, Purplle, and Tira. Each record describes a single marketing campaign and contains the following attributes.

- Identifiers and metadata: campaign ID, brand source, campaign type, channel used, date, duration (in days)
- Audience attributes: customer segment, target audience, language
- Performance metrics: impressions, clicks, leads, conversions, revenue, acquisition cost, ROI, and engagement score

We built an end-to-end analytics pipeline using a medallion architecture with Bronze, Silver, and Gold layers. The pipeline supports exploratory data analysis and prepares a modeling-ready dataset. We then train a binary classification model to predict whether a campaign will fall in the top 25% by ROI using information available before the campaign launches.

Our core business question is:

Given only pre-campaign attributes (channel, campaign type, customer segment, language, duration, and timing), can we predict which campaigns will be high-ROI?

This could help marketing teams allocate budget before campaigns launch.

Tools and environment:

We used PySpark 4.0.1 for distributed processing, Spark SQL for analysis, Spark MLlib for machine learning pipelines, and MLflow for experiment tracking. We also used Pandas for lightweight inspection and Matplotlib for visualizations.

Phase 2: Data Ingestion & Preprocessing

In this phase, we ingested three campaign datasets from Nykaa, Purplle, and Tira and prepared a clean, standardized silver table for analysis. Each brand contributed 55,555 campaign records for a total of 166,665 rows spanning the period from July 2024 to June 2025. The raw data was distributed across three separate CSV files, one per brand, with

slightly inconsistent schemas and mixed date formats that required preprocessing before analysis.

File	Brand
data/nykaa_campaign_data.csv	Nykaa
data/purplle_campaign_data.csv	Purplle
data/tira_campaign_data.csv	Tira

Dataset Schema (Raw):

Column	Type	Description
campaign_id	string	Unique campaign identifier
campaign_type	string	Email, Social Media, Influencer, Paid Ads, SEO
channel_used	string	One or more channels (comma-separated)
impressions	long	Total ad impressions
clicks	long	Total ad clicks
leads	long	Leads generated
conversions	long	Conversions achieved
acquisition_cost	double	Total spend on campaign
revenue	long	Revenue attributed to campaign
roi	double	Return on investment
engagement_score	double	Composite engagement metric
duration	long	Campaign duration in days
customer_segment	string	Premium Shoppers, College Students, Youth, etc.
target_audience	string	Audience targeting label
language	string	Language of campaign (e.g., Bengali, Tamil)

date	timestamp	Campaign date

a. Data Ingestion Pipeline

1) Schema Normalization: Column names were converted to lowercase, and non-alphanumeric characters were replaced with underscores to ensure consistency across all three CSV sources.

2) Data Parsing: Two date formats were handled for consistency, including "dd-MM-yyyy" and "MM/dd/yyyy".

3) Brand Tagging: A new column called "brand_source" was added to each row to preserve the original data source after performing the union operation.

4) Schema Alignment: Missing columns across the DataFrames were filled with null values before performing the union using unionByName.

5) Deduplication: Duplicate rows were removed using the dropDuplicates() function to ensure data quality.

6) Spark Session - Spark was initialized with 8GB driver memory and 8 shuffle partitions to support efficient local execution.

Data Ingestion Result:

Metric	Value
Total rows after ingestion	166,665
Number of brands	3
Raw columns	16

b. Medallion Architecture and Silver Layer Design

To organize the data pipeline, we implemented a medallion architecture consisting of Bronze, Silver, and Gold layers. The Bronze layer contains the raw ingested data with minimal transformation. In this phase, we focused on creating the Silver layer, which contains cleaned and standardized data ready for analysis and modeling. The Gold layer will be created later to support machine learning and advanced analytics.

c. Data Cleaning and Quality Checks

Data cleaning steps were applied to improve data consistency and quality.

Step	Action
Null strings	Replaced with "Unknown"
Whitespace	Trimmed from all string columns
Negative numeric values	Replaced with null for non-negative columns (clicks, impressions, conversions, revenue, etc.)
ROI negatives	Retained (negative ROI is a valid business outcome)

These cleaning steps produce a standardized Silver dataset that supports reliable exploratory analysis and feature engineering in later phases.

d. Feature Engineering for the Silver Layer

To better capture campaign performance and funnel efficiency, we created eight KPI features. These derived variables represent different stages of the marketing funnel and provide interpretable indicators of campaign performance. The KPI metrics support exploratory analysis and help summarize campaign effectiveness across the dataset.

The engineered KPI metrics are summarized below.

KPI	Formula	Business Meaning
ctr	clicks / impressions	Click-through rate; ad creative effectiveness
lead_rate	leads / clicks	Landing page / top-of-funnel conversion
lead_to_conv	conversions / leads	Lead qualification quality
click_to_conv	conversions / clicks	Full funnel conversion rate
cpc	acquisition_cost / clicks	Cost per click
cpa	acquisition_cost / conversions	Cost per acquisition
roas	revenue / acquisition_cost	Return on ad spend
rpm	(revenue × 1000) / impressions	Revenue per thousand impressions

To maintain numerical stability, safe division was applied during feature creation. If the denominator was less than or equal to zero, the resulting value was set to null. This prevents invalid ratios from being generated.

For funnel related ratios such as ctr, lead_rate, lead_to_conv, and

click_to_conv, values were clipped to the range between 0 and 1 to keep the metrics within logical bounds.

e. Silver Layer Output

After completing the data cleaning and feature engineering steps, the resulting Silver dataset contained 166,665 rows and 25 columns. The dataset was saved in Parquet format at 'data/marketing_silver_parquet' to improve storage efficiency and support faster reads during later stages of the pipeline.

In addition, the Silver dataset was registered as a Spark SQL temporary view named marketing_silver. This allowed us to run SQL queries directly on the cleaned dataset during exploratory data analysis and validation steps in the following phase.

Phase 3: Exploratory Data Analysis (EDA)

a. Summary Statistics

Key summary metrics are shown below.

Summary Statistics:

Metric	Value
Total campaigns	166,665
Average impressions	55,060.85
Average clicks	4,682.37
Average conversions	1,029.09
Average CTR	8.50%
Average ROI	2.69

Numeric Distribution:

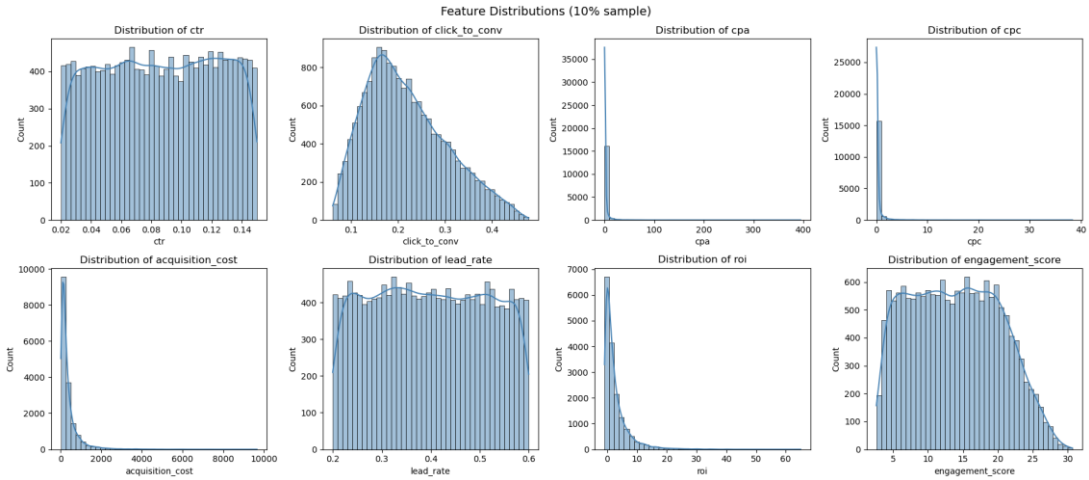
Column	Min	25th pct	Median	Mean	75th pct	Max
acquisition_cost	8.18	106.71	208.47	376.09	427.40	15,473.16
clicks	202	2,108	3,903	4,682	6,687	14,944

conversions	17	401	776	1,029	1,403	6,686
duration	5	11	17	17.49	24	30
engagement_score	2.56	8.38	13.59	13.77	18.79	30.99
impressions	10,001	32,561	55,102	55,061	77,565	100,000
leads	48	779	1,475	1,872	2,598	8,876
revenue	3,895	177,660	359,172	513,907	684,632	4,579,910
roi	-0.99	0.04	1.23	2.69	3.58	79.30
ctr	0.020	0.053	0.085	0.085	0.118	0.150
lead_rate	0.198	0.299	0.399	0.399	0.499	0.600
lead_to_conv	0.288	0.424	0.550	0.550	0.674	0.800
click_to_conv	0.059	0.152	0.205	0.219	0.277	0.479
cpc	0.001	0.017	0.052	0.320	0.189	59.51
cpa	0.001	0.078	0.264	2.076	1.027	870.73
roas	0.259	444.52	1,761.82	6,439.07	6,112.35	504,395.37
rpm	299.73	4,221.18	7,457.84	9,331.38	12,513.86	51,321.10

b. Feature Distribution Analysis

To better understand the distribution of key campaign metrics, we visualized several numeric features using histograms based on a 10 percent sample of the dataset. Sampling was used to reduce memory usage while preserving the overall distribution patterns.

The distributions reveal several important patterns.



First, ROI shows a strong right-skewed distribution. Most campaigns generate modest returns, while a small number of campaigns achieve extremely high ROI values, indicating a small number of exceptionally high-performing campaigns pulling the average upward. At the extreme tail, some campaigns achieve an ROI as high as 79.3. This skewed distribution suggests that a small number of campaigns generate disproportionately large returns, which can significantly influence average performance metrics. As a result, predicting high-performing campaigns becomes more challenging because a few extreme outcomes dominate the variance in ROI values. This pattern also reflects the inherent uncertainty of marketing performance, where a limited number of campaigns may achieve exceptionally strong engagement or conversion outcomes.

Second, cost related variables such as acquisition cost, CPC, and CPA are also heavily right-skewed. This suggests that while most campaigns operate within moderate cost ranges, a small number incur significantly higher marketing expenses.

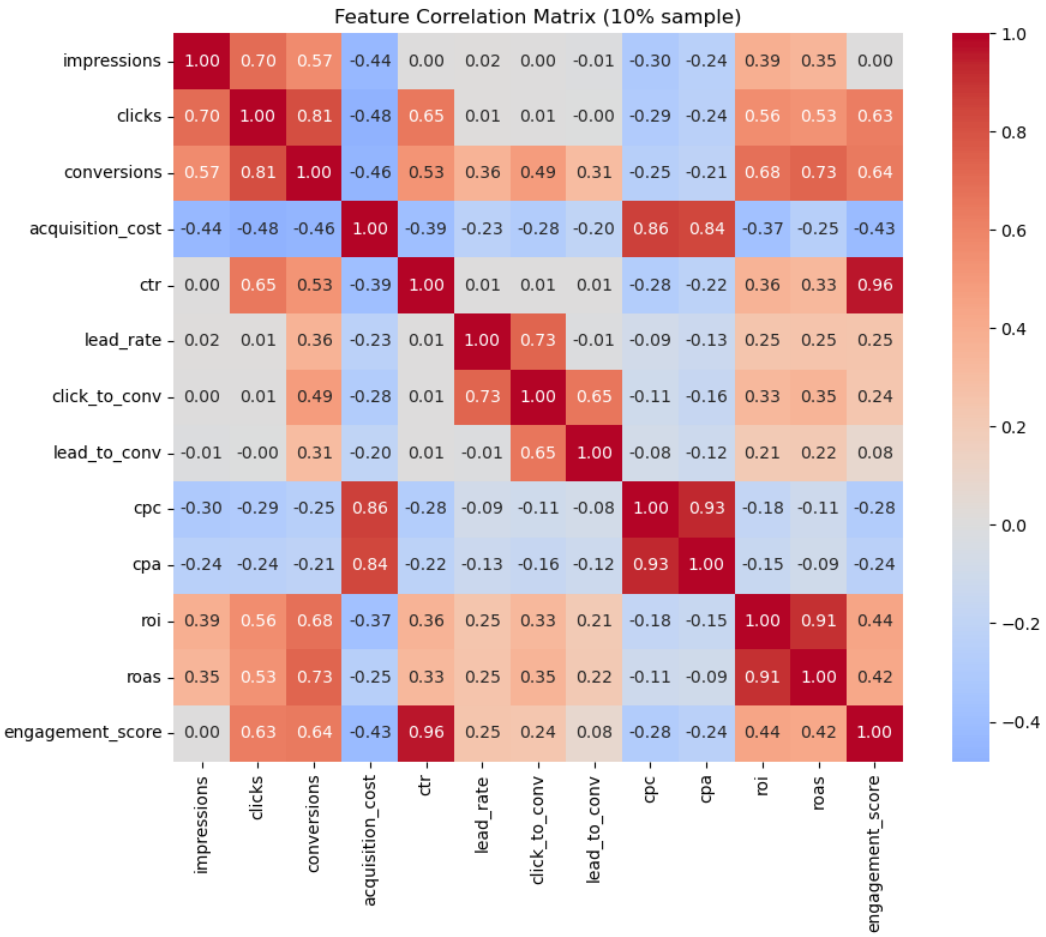
Third, funnel conversion metrics such as CTR, lead_rate, and click_to_conv appear relatively stable within bounded ranges. These metrics represent ratios derived from campaign interactions and therefore remain within logical limits.

Finally, engagement score shows a relatively smooth distribution across its range. This suggests that audience engagement varies gradually rather than being concentrated in extreme values.

Overall, these distributions indicate that campaign outcomes vary widely, particularly for financial return metrics such as ROI and revenue efficiency.

c. Correlation Analysis

To examine relationships between campaign variables, we computed a correlation matrix using a 10 percent sample of the dataset. The heatmap highlights several meaningful relationships between marketing metrics.



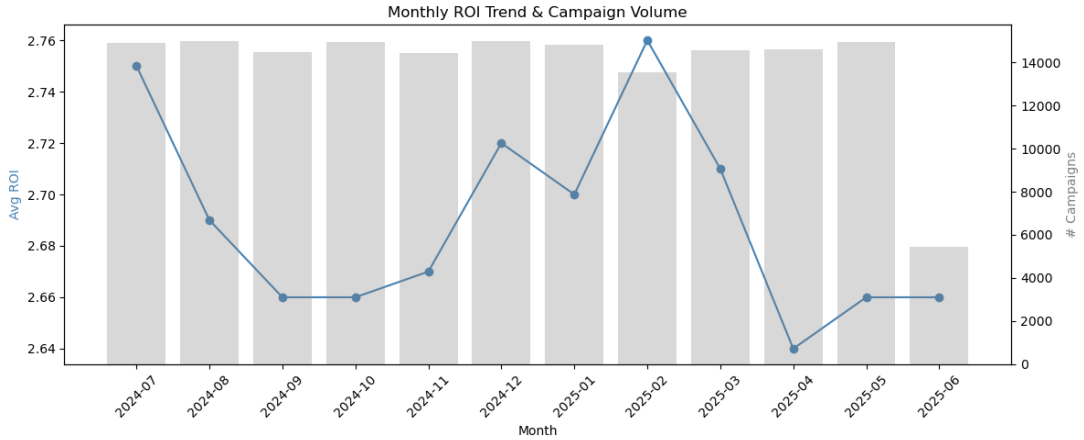
Strong positive relationships appear between impressions, clicks, and conversions. This is expected because campaigns that reach more users typically generate more interactions and conversions.

Several derived funnel metrics are also correlated. For example, lead_rate and click_to_conv show a relatively strong positive relationship, reflecting their shared dependence on conversion outcomes. Cost related variables such as CPC and CPA are strongly correlated with acquisition cost. This is expected because these ratios are calculated directly from cost values. ROI shows moderate positive correlation with conversions, clicks, and engagement score. This suggests that campaigns generating higher user interaction tend to produce higher financial returns.

These correlation patterns help identify relationships between engagement metrics, cost efficiency, and campaign outcomes. In particular, several post-campaign performance metrics such as ROAS, CPA, and conversion-related variables show strong correlations with ROI. Because these variables are directly derived from campaign outcomes, including them in a predictive model would introduce information that is only available after the campaign has already occurred.

d. Campaign Patterns

To analyze patterns in campaign performance, we examined average ROI and campaign volume by month.



Campaign activity remained relatively stable throughout the year, with similar campaign volumes across most months. However, average ROI fluctuates across months.

ROI appears higher in certain periods such as early 2025, while some months show slightly lower performance levels. These fluctuations suggest that marketing effectiveness may vary over time due to seasonality, promotional cycles, or consumer demand changes.

Understanding these temporal patterns helps contextualize campaign performance and may inform future marketing scheduling strategies.

e. Campaign Type Performance

Campaign performance was also examined by campaign type.

campaign_type	n_campaigns	avg_roi	avg_ctr	avg_conversion	avg_cpa
Paid Ads	33283	2.71	0.0851	0.2195	2.02
Social Media	33179	2.71	0.0853	0.2187	2.05
Email	33493	2.69	0.0848	0.22	2.11
SEO	33207	2.69	0.0847	0.22	2.09
Influencer	33503	2.65	0.0853	0.2191	2.1

The dataset includes five primary campaign categories including Paid Ads, Social Media, Email, SEO, and Influencer campaigns. Campaign volume is relatively evenly distributed across these categories.

Average ROI across campaign types remains relatively similar, ranging between approximately 2.65 and 2.71. CTR and conversion rates also remain consistent across types.

These results suggest that no single campaign type overwhelmingly dominates in terms of ROI. Instead, campaign effectiveness may depend more strongly on other factors such as audience targeting, timing, and channel mix.

f. Channel Performance

Since campaigns can use multiple marketing channels, we expanded the channel_used column to analyze individual channel performance.

channel	n	avg_roi	avg_ctr
Email	55590	2.7298	0.0853
Instagram	55599	2.7223	0.0853
Google	55576	2.6887	0.0847
Facebook	55578	2.6798	0.0848
WhatsApp	55455	2.6776	0.0852
YouTube	55583	2.6696	0.0848

Campaigns were distributed across six primary marketing channels including Email, Instagram, Google, Facebook, WhatsApp, and YouTube.

Average ROI across channels is relatively similar, although Email and Instagram show slightly higher average returns compared with other channels. CTR values remain nearly identical across channels, suggesting that user engagement levels are consistent across marketing platforms.

These results indicate that while channel selection may influence campaign outcomes slightly, performance differences across channels are relatively modest. Overall, the exploratory analysis highlights that campaign outcomes vary widely across the dataset, particularly for financial performance metrics such as ROI. These patterns suggest that predicting high-performing campaigns using only structural attributes may be difficult, which motivates the predictive modeling approach explored in the following section.

Phase 4: Predictive Modeling using Spark MLlib

a. Modeling objective and design

The goal of Phase 4 is to predict whether a campaign will be a high ROI campaign using only information available before the campaign launches. The prediction task is framed as a binary classification problem where campaigns in the top 25 percent of ROI outcomes are labeled as high performing campaigns.

To follow the medallion architecture, Phase 4 reads directly from the saved Silver Parquet dataset instead of re-running the entire ETL pipeline. This design improves efficiency and ensures that modeling is performed on the standardized Silver layer created in the earlier phases.

b. Target definition and leakage prevention

Before training the models, we defined the binary target variable that identifies high ROI campaigns.

To prevent label leakage, the ROI threshold used to define the top 25 percent was computed using only the training data. This ensures that the test data does not influence the definition of the target variable. The 75th percentile ROI in the training set was approximately 3.59. Campaigns with ROI greater than or equal to this

threshold were labeled as high performing campaigns.

The resulting class distributions were as follows.

Train set:

Label	Meaning	Count	Share
1	High-performing (top 25% ROI)	33,459	25.0%
0	Standard ROI	100,201	75.0%

Test set:

Label	Meaning	Count	Share
1	High-performing (top 25% ROI)	8,147	24.7%
0	Standard ROI	24,858	75.3%

c. Feature construction

The modeling dataset includes only pre-campaign attributes to ensure that predictions rely on information available before a campaign begins.

We created additional temporal features 'year', 'month', 'dayofweek' derived from the campaign start date. In addition, the channel_used column contains multiple channels in a single string. To represent this information numerically, we created binary indicator columns for each channel.

The final feature set includes:

Feature Type	Features
Categorical (OHE)	brand_source, campaign_type, customer_segment, language, year, month, dayofweek
Numeric	duration
Channel flags (binary)	ch_facebook, ch_whatsapp, ch_youtube, ch_google, ch_email, ch_instagram

Post campaign variables such as clicks, conversions, cost, revenue, and ROI were excluded from the feature set to prevent data leakage.

d. Train/Test Split

The dataset was split into training and test sets using an 80 percent to 20 percent ratio.

- Training set size: 133,660 campaigns
- Test set size: 33,005 campaigns

Splitting was performed before label creation to ensure that the ROI threshold used to define the label was derived exclusively from the training data.

e. Machine Learning Pipeline

We implemented a Spark ML pipeline to ensure consistent preprocessing and feature transformation.

Pipeline: StringIndexer → OneHotEncoder → VectorAssembler → Classification

- StringIndexer: to convert categorical variables into indexed values
- OneHotEncoder: to represent categorical variables as binary vectors
- VectorAssembler: to combine all numeric, encoded categorical, and channel indicator features into a single feature vector
- A classification model as the final stage

This pipeline structure ensures that the same preprocessing steps are applied consistently during both training and prediction.

f. Model training and hyperparameter tuning

Three classification algorithms were trained using Spark MLlib. We selected three commonly used classification algorithms for structured tabular data: Logistic Regression, Random Forest, and Gradient Boosted Trees. These models represent both linear and non-linear approaches and are widely supported in the Spark MLlib ecosystem. Hyperparameters for each model were tuned using three-fold cross validation with area under the ROC curve as the primary evaluation metric.

Model	Hyperparameter Grid
Logistic Regression	regParam ∈ {0.01, 0.1}, elasticNetParam ∈ {0.0, 0.5}, maxIter=50
Random Forest	numTrees ∈ {50, 100}, maxDepth ∈ {3, 5}
GBT Classifier	maxDepth ∈ {3, 5}, stepSize ∈ {0.05, 0.1}, maxIter=20

g. Model evaluation

Models were evaluated on the test dataset using the following metrics, and the results are summarized below.

Model	AUC	Precision	Recall	F1
Logistic Regression	0.4989	0.5672	0.7532	0.6471
Random Forest	0.4955	0.5672	0.7532	0.6471
GBT Classifier	0.4995	0.5672	0.7532	0.6471

→ Best model: GBT Classifier

Because the dataset is moderately imbalanced, predicting the majority class alone would already achieve relatively high accuracy. Therefore, metrics such as AUC, precision, recall, and F1-score provide a more informative evaluation of model performance than accuracy alone.

AUC was used as the primary metric during cross-validation to compare model performance, while precision, recall, and F1-score were evaluated on the test set to provide a more complete view of classification performance. All models achieved AUC values close to 0.50, indicating performance near random classification. Precision, recall, and F1 scores are nearly identical across the three models, suggesting that the models primarily predict the majority class.

This result indicates that pre-campaign attributes capture structural characteristics of marketing campaigns but do not reflect real-time audience engagement or behavioral responses. Many of the strongest predictive signals for marketing performance likely emerge during campaign execution through interactions such as clicks, engagement, and conversions.

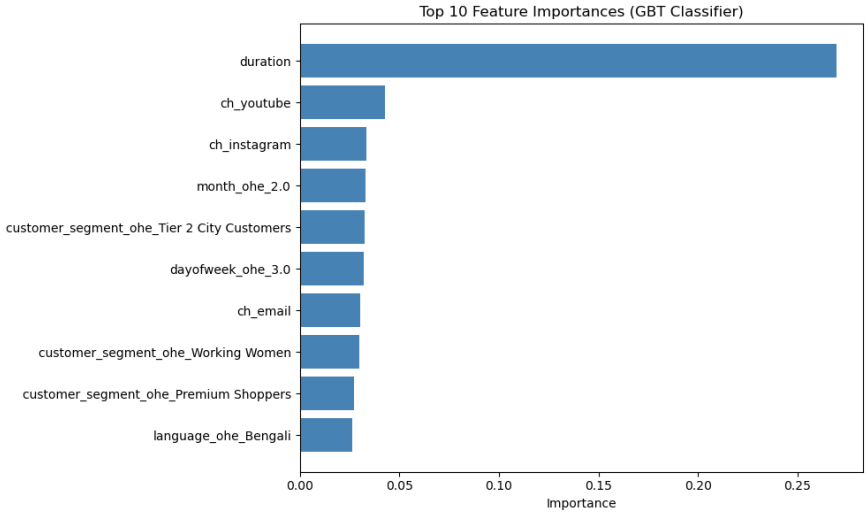
h. Feature importance

Although overall predictive performance is limited, we examined feature importance from the model with the highest AUC (GBT classifier) to better understand which pre-campaign attributes contribute most to the model's decision process.

Top 10 Important Features:

Rank	Feature	Coefficient Magnitude
1	duration	0.269344

2	ch_youtube	0.042842
3	ch_instagram	0.033615
4	month_ohe_2.0	0.032838
5	customer_segment_ohe_Tier 2 City Customers	0.032412
6	dayofweek_ohe_3.0	0.031976
7	ch_email	0.030319
8	customer_segment_ohe_Working Women	0.029896
9	customer_segment_ohe_Premium Shoppers	0.027310
10	language_ohe_Bengali	0.026343



Feature importance analysis from the GBT model shows that campaign duration is the most influential feature, followed by several channel indicators and temporal features. These results suggest that structural campaign attributes such as duration, channel mix, and launch timing contribute modest signal when predicting high ROI campaigns. The feature importance plot visualizes the relative contribution of each feature to the GBT model's decision process.

Because overall model performance is close to baseline, these feature importance values should be interpreted with caution. However, they still provide useful indications of which pre-campaign attributes are most influential in the model's decision process.

i. Key Takeaway

The modeling results suggest that pre-campaign attributes alone are insufficient to reliably predict whether a campaign will fall within the top 25 percent

of ROI outcomes.

This finding highlights the importance of post-campaign engagement signals such as clicks, conversions, and cost efficiency (ROAS) when predicting marketing performance. Future work could incorporate additional pre-campaign planning variables such as audience size, promotional strategy, or historical campaign performance to improve predictive power.

Phase 5: MLOps Best Practices

a. Experiment tracking with MLflow

To ensure reproducibility and systematic experiment management, we used MLflow to track model training runs. An experiment named Campaign_ROI_Classification was created to log all model runs. Each run recorded the model type, hyperparameters, evaluation metrics, and the serialized Spark model artifact.

The modeling workflow itself was implemented as a Spark ML pipeline, which automates preprocessing, feature transformation, and model training in a single executable workflow. When combined with MLflow tracking, this allows each pipeline execution to be logged as a reproducible experiment run.

The following models were trained and logged.

Run Name	Model	AUC	F1
LR_Pipeline_CV	Logistic Regression	0.4989	0.6471
RF_Pipeline_CV	Random Forest	0.4955	0.6471
GBT_Pipeline_CV	GBT Classifier	0.4995	0.6471

This experiment tracking allows all model configurations and results to be stored in a centralized log, making it easier to compare models and reproduce results.

b. Model registry and reproducibility

The best performing model based on AUC was the GBT classifier. This model was registered in the MLflow Model Registry and then loaded again for automated inference.

The loaded model produced the same evaluation metrics as the original training run.

Metric	Value

Model	GBT Classifier
AUC	0.4995
Precision	0.5672
Recall	0.7532
F1	0.6471

Because the metrics remained identical after reloading the model artifact, this confirms that the entire pipeline can be serialized and reproduced reliably. Using MLflow artifacts also enables efficient deployment. Any previously trained model can be reloaded and applied to new campaign data without retraining.

Phase 6: Insights and Business Recommendations

a. Key findings from the predictive models

The predictive modeling results show that pre-campaign attributes alone provide limited ability to distinguish high ROI campaigns. All three models achieved AUC values close to 0.50 and produced nearly identical Precision, Recall, and F1 scores. This indicates that the models largely default to predicting the majority class. In practice, this means that campaign characteristics available before launch do not contain strong predictive signals for identifying campaigns that will fall within the top 25% of ROI outcomes.

Despite this limitation, the analysis still reveals several structural patterns that may help guide marketing strategy and experimentation. Feature importance analysis from the GBT model highlights several attributes that appear more frequently among higher-ROI campaigns.

- Campaign duration is the most influential structural feature in the model. Its importance score is significantly larger than other variables, suggesting that campaign duration may be associated with the accumulation of impressions and conversions over time.
- Channel indicators, particularly YouTube and Instagram, also appear among the most important features. This suggests that campaigns incorporating video and social channels may show different engagement patterns compared with campaigns relying on other channels.
- Temporal features such as month and day of week also appear among the top features in the model, indicating that campaign launch timing may have some relationship with campaign performance.
- Several customer segments also appear among the more influential variables,

suggesting that targeting strategies may play a role in campaign outcomes across different audience groups.

b. Business recommendations

Although the predictive model does not strongly separate high-ROI campaigns from others, the analysis still provides several directional insights that marketing teams can explore through controlled experimentation.

i. Investigate the role of campaign duration

Among all structural variables, campaign duration shows the strongest influence in the model. In the GBT classifier, duration has the highest feature importance (0.27), far larger than other variables. Marketing teams should analyze whether longer campaigns consistently produce higher ROI or whether this relationship is driven by correlated factors such as campaign budget, audience size, or campaign type. If a causal relationship exists, this insight could directly influence campaign planning and budget allocation strategies.

ii. Experiment with channel combinations

YouTube and Instagram appear as the strongest channel indicators in the model, ranking second and third in feature importance. Campaigns that include these platforms appear somewhat more frequently among higher ROI outcomes. Marketing teams may consider testing campaign strategies that incorporate these channels more frequently.

iii. Evaluate campaign launch timing

Temporal features also play a noticeable role in the model's decision patterns. February and Wednesday appear as notable split points in the GBT classifier. This pattern may reflect seasonal promotions such as Valentine's Day campaigns, post-holiday purchasing cycles, or mid-week engagement patterns. Testing different launch schedules could help determine whether campaign timing influences engagement and conversion behavior.

iv. Explore emerging customer segments

Customer segments such as Tier 2 City Customers and Working Women appear among the more influential variables in the model alongside Premium Shoppers. These segments may represent emerging growth audiences within the Indian beauty e-commerce market. Further segmentation analysis could help determine whether campaigns targeting these audiences consistently produce stronger ROI outcomes.

v. Regional targeting

Regional language targeting shows only a marginal signal in the model. Bengali appears at rank 10 in the GBT feature importance list, suggesting that regional campaigns may behave somewhat differently in aggregate. However, the effect size is small and should not be treated as directly actionable without more targeted regional analysis.

c. Challenges and possible improvements

Several challenges emerged during the project.

First, restricting the model to pre-campaign attributes limits the available predictive signal. Engagement metrics that emerge during campaign execution, such as CTR, engagement score, and ROAS, would likely provide stronger predictive information but cannot be used in a pre-launch prediction system without introducing data leakage.

Second, the dataset shows moderate class imbalance, with approximately 75% of campaigns labeled as standard ROI. While this imbalance may encourage majority-class predictions, the main challenge appears to be the relatively weak signal contained in pre-campaign attributes rather than the class distribution itself.

Third, computational constraints limited the scope of hyperparameter tuning. Running the pipeline on a larger distributed cluster would allow a broader search of model configurations.

Future improvements could include applying class balancing techniques such as class weights, incorporating external signals such as seasonal events or market indicators, creating interaction features between marketing channels, and testing alternative algorithms such as LightGBM or XGBoost. Additionally, a regression model that predicts the continuous ROI value could complement the binary classification approach.

The MLflow pipeline developed in this project could also serve as a preliminary screening tool for new campaign plans. Although the current model provides limited predictive power, it establishes a reproducible baseline that can be improved as more campaign data becomes available. Future versions of the system could incorporate early in-campaign signals such as CTR or engagement metrics after a campaign has been running for several days, creating a two-stage scoring approach that combines pre-launch planning features with early performance indicators.

Summary

This project developed an end-to-end marketing analytics pipeline covering data ingestion, exploratory analysis, predictive modeling, and MLOps deployment for 166,665 marketing campaigns across Nykaa, Purplle, and Tira.

The predictive modeling results show that pre-campaign attributes alone provide limited ability to identify high-ROI campaigns. Logistic Regression, Random Forest, and Gradient Boosted Tree models all achieved AUC values close to 0.50, indicating near-random predictive performance. This suggests that the most predictive signal for marketing performance emerges during campaign execution rather than from pre-campaign

characteristics.

Despite this limitation, feature importance analysis highlights several structural attributes that may influence campaign outcomes, including campaign duration, channel selection, timing, and customer segmentation. These insights provide useful starting points for marketing experimentation.

The project also demonstrates a reproducible MLOps-ready workflow using Spark ML pipelines and MLflow experiment tracking. Future improvements could incorporate richer signals, additional feature engineering, and alternative modeling approaches to improve predictive performance.